



Android Release & Google Play Guide

From a signed **.apk** / **.aab** to internal testing, closed testing and a public Production release on Google Play

React Native CLI 0.81

Gradle

Upload key

Play Console

Internal testing

Updates

App: **Shopvra** · Package: **com.ritikji.clientEcommerce** · Current release: **1.0.1 (versionCode 2)**

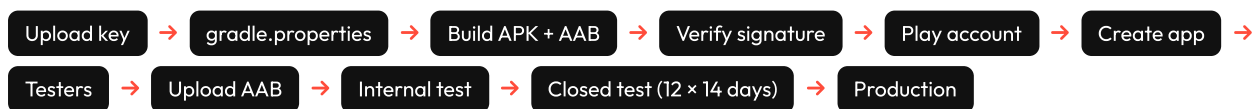
Backend: <https://api.shopvra.space> · Web: <https://shopvra.space> · Prepared October 2026

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The whole journey at a glance



APK vs AAB - which file is for what?

	.apk (Android Package)	.aab (Android App Bundle)
Used for	Installing directly on a phone (USB, WhatsApp, Drive, email)	Uploading to Google Play - Play builds optimised APKs per device from it
Play Store accepts it?	✗ No (new apps must use AAB)	✓ Yes - required
Size (Shopvra 1.0.0)	67 MB (contains every CPU type)	47 MB upload → 13.2 MB download for users
Gradle task	<code>assembleRelease</code>	<code>bundleRelease</code>



After internal testing - your step-by-step checklist

Internal testing of **1.0.0** is done (live for internal testers on Oct 10, 2026). Work through these steps **in order** to publish Shopvra on Google Play for everyone. Tick each box when it is done; the last column points to the detailed instructions.

✓	Step	Details
Phase 1 - Put the new features live (today)		
☐	Deploy the backend: <code>npm run build:lambda</code> → AWS Console → Lambda → Shopvra function → Upload <code>lambda.zip</code> . Check <code>DELETE https://api.shopvra.space/api/auth/me</code> returns “Not authorized”, not 404.	Section 13 (A)
☐	Deploy the website: <code>npm run build</code> → upload <code>dist/*</code> to the shopvra.space S3 bucket → CloudFront invalidation <code>/*</code> .	Section 13 (A)
☐	Open shopvra.space/privacy and shopvra.space/delete-account in a private window - both must load.	Section 13 (A)
☐	Make sure support@shopvra.space is a working inbox (used in the policy and the store listing).	Section 15
☐	Upload 1.0.1 (versionCode 2) to Internal testing (adds Delete account + Privacy Policy link).	Section 13 (B)
☐	On your phone: update from Play → sign in with a test account → Profile → Delete account works.	Section 12
Phase 2 - Complete the Play Console setup (1-2 hours)		
☐	Create a reviewer demo account in the app (not the admin) and note its email + password.	Section 14
☐	App content: Privacy policy, App access, Ads, Content rating, Target audience, News, Advertising ID, Government, Financial features, Health.	Section 14
☐	Data safety form incl. Delete account URL <code>https://shopvra.space/delete-account</code> .	Section 14
☐	Store listing: name, short + full description (ready text), category Shopping, contact email, website.	Section 15
☐	Graphics: 512×512 icon, 1024×500 feature graphic, 2-8 phone screenshots.	Section 15
☐	Dashboard → every “Set up your app” task shows a green tick.	Section 18
Phase 3 - Closed testing (14+ days)		
☐	Closed testing track → countries → testers list / Google Group with 15-20 people (minimum 12) → feedback email.	Section 16
☐	Create closed release → Add from library: 2 (1.0.1) → Save and publish → wait for Google review.	Section 16
☐	Send the opt-in link ; every tester joins and installs from Play.	Section 16
☐	Keep ≥ 12 testers opted in for 14 consecutive days ; collect feedback, fix bugs (new builds: raise versionCode).	Sections 16, 20
Phase 4 - Production (about 1-2 weeks)		
☐	Dashboard → Apply for production → answer the questions about your test → wait for the email (≈ 7 days).	Section 17
☐	Production → countries → Create new release (from library) → user release notes.	Section 18
☐	Staged rollout 20% → Send for review → when live, watch Android vitals → raise to 50% → 100%.	Section 18
☐	App is public at play.google.com/store/apps/details?id=com.ritikji.clientEcommerce - share the link 🎉	Section 18
Phase 5 - After launch (ongoing)		

☐	Check crashes/ANRs weekly, reply to reviews, answer policy emails, keep api.shopvra.space and the website online.	Section 19
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☐	Every update: raise <code>versionCode</code> + <code>versionName</code> → build → upload to the track → roll out.	Section 20
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1 Overview & key facts

Item	Value
App name (phone + Play Store)	Shopvra
Package name / applicationId	<code>com.ritikji.clientEcommerce</code> - permanent , can never change on Play
Version	<code>versionCode 2</code> , <code>versionName "1.0.1"</code> (1.0.0 / code 1 was the first internal test) in <code>android/app/build.gradle</code>
Project folder	<code>frontend/mobile_ecommerce_app/cli</code>
Release APK	<code>cli/android/app/build/outputs/apk/release/app-release.apk</code>
Release AAB	<code>cli/android/app/build/outputs/bundle/release/app-release.aab</code>
Upload keystore	<code>/Users/apple/keys/shopvra-upload.keystore</code> (alias <code>upload</code>)
Signing settings	<code>~/ .gradle/gradle.properties</code> (outside the project, never committed)
Certificate owner	CN=Ritik Sharma, O=Shopvra, L=India, C=IN
Upload cert SHA-256	<code>30:07:B9:39:C7:99:54:EE:C6:7F:0E:D1:9E:99:F8:D1:DF:FC:F3:32:32:69:4F:DC:EB:F7:00:5F:E8:F9:91:D5</code>
Backend used by release builds	<code>https://api.shopvra.space</code> (set in <code>src/config/env.ts</code>)
Min / target Android	API 24 (Android 7.0) / API 36

How signing works with Google Play

You sign the AAB with your **upload key**. With **Play App Signing** (enabled on the first upload) Google re-signs the app with its own **app signing key** before delivering it to users. If you ever lose the upload key, Google can reset it - but it takes days, so keep it backed up.

2 Prerequisites

- macOS with **Android Studio** / Android SDK (gives `adb`, `apksigner`) and **JDK 17** (gives `keytool`).
- Project dependencies installed: `cd frontend/mobile_ecommerce_app/cli && npm install`.
- A Google account for Play Console and a card for the one-time **US \$25** registration fee.
- An Android phone with **USB debugging** enabled (only needed to test the APK).

Quick health check before every release

`npm run validate` (TypeScript + ESLint + Jest) should pass.

3 Create the Play Store upload key (one time only)

Run in Terminal. It asks for a password – choose a strong one and **use the same password** when asked for the key password.

```
mkdir -p ~/keys && keytool -genkeypair -v -storetype PKCS12 \  
-keystore ~/keys/shopvra-upload.keystore -alias upload \  
-keyalg RSA -keysize 2048 -validity 10000 \  
-dname "CN=Ritik Sharma, O=Shopvra, L=India, C=IN"
```

Flag	Meaning
-keystore	File that stores the key (keep it outside the git project)
-alias upload	Name of the key inside the keystore
-validity 10000	Valid for ~27 years (Google requires validity past 2033)
-dname	Owner details printed in the certificate

Never lose this file or its password

Every future update of Shopvra must be signed with the same upload key. Back up `~/keys/shopvra-upload.keystore` and the password (see section 17).

4 Tell Gradle about the key (one time only)

Create / open the global Gradle settings file (it lives in your home folder, not in the project):

```
touch ~/.gradle/gradle.properties && open -e ~/.gradle/gradle.properties
```

Add these 4 lines (replace `YOUR_PASSWORD`) and save:

```
MYAPP_UPLOAD_STORE_FILE=/Users/apple/keys/shopvra-upload.keystore  
MYAPP_UPLOAD_KEY_ALIAS=upload  
MYAPP_UPLOAD_STORE_PASSWORD=YOUR_PASSWORD  
MYAPP_UPLOAD_KEY_PASSWORD=YOUR_PASSWORD
```

```
chmod 600 ~/.gradle/gradle.properties ~/keys/shopvra-upload.keystore # only your user can read them
```

How the project uses it

`android/app/build.gradle` reads these properties for the `release` signing config. If they are missing, `assembleRelease` falls back to the debug key (fine for testing on a phone) and `bundleRelease` **stops with an error** – so a debug-signed AAB can never be uploaded by mistake.

Tip: `open -e` cannot open a file that does not exist – that is why the command above runs `touch` first.

5 Build the release .apk and .aab

```
cd frontend/mobile_ecommerce_app/cli/android
./gradlew assembleRelease bundleRelease
```

Takes 3-10 minutes. Outputs:

File	Path (inside <code>cli/android</code>)
APK	<code>app/build/outputs/apk/release/app-release.apk</code>
AAB	<code>app/build/outputs/bundle/release/app-release.aab</code>

Or from the `cli` folder with npm scripts: `npm run build:apk` and `npm run build:aab`.

Do not run `clean` in the same command

`./gradlew clean assembleRelease` fails with *“react-native-worklets ... prefab ... is not readable”*. If you need a clean build, run `./gradlew clean` first, then the build as a separate command.

6 Verify the files are signed correctly

```
BT=$(ls -d ~/Library/Android/sdk/build-tools/* | sort -V | tail -1)
$BT/apksigner verify --print-certs app/build/outputs/apk/release/app-release.apk | grep DN
keytool -printcert -jarfile app/build/outputs/bundle/release/app-release.aab | grep Owner
$BT/aapt2 dump badging app/build/outputs/apk/release/app-release.apk | grep -E "^package|application-label"
```

Expected output:

```
certificate DN: CN=Ritik Sharma, O=Shopvra, L=India, C=IN           # APK - your upload key
Owner: CN=Ritik Sharma, O=Shopvra, L=India, C=IN                 # AAB - your upload key
package: name='com.ritikji.clientEcommerce' versionCode='1' versionName='1.0.0'
application-label: 'Shopvra'
```

If you see `CN=Android Debug`, Gradle did not find the properties from section 4.

7 Install the .apk on a phone

1. On the phone: **Settings** → **About phone** → tap **OS version** 7 times → **Developer options** → **USB debugging** ON (Xiaomi/POCO: also **Install via USB**).
2. Connect the USB cable and tap **Allow** on the “Allow USB debugging?” prompt.

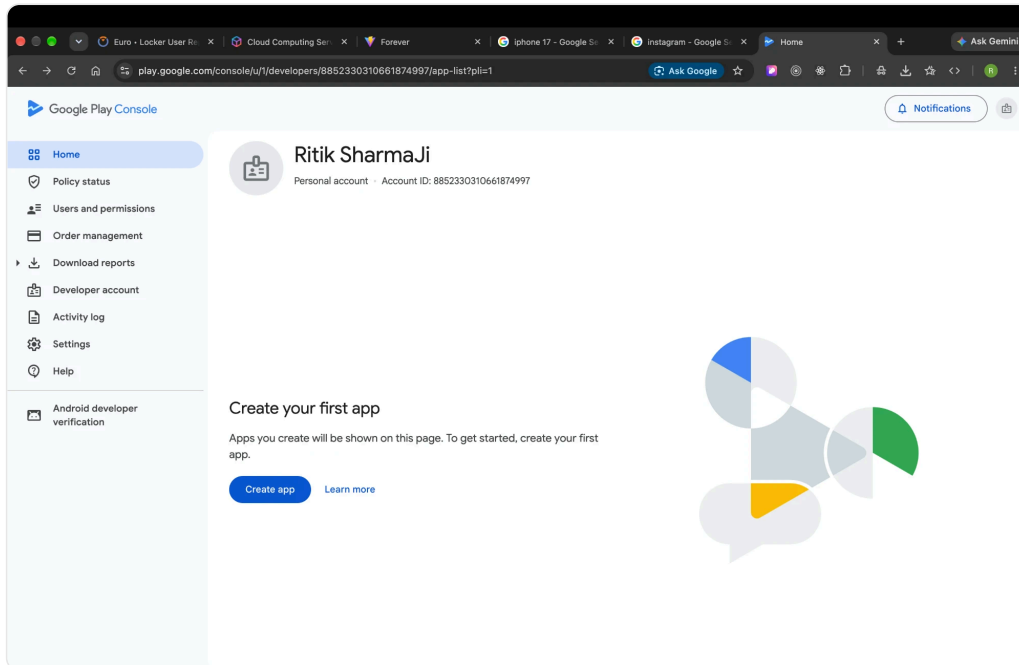
```
adb devices                # phone must show as "device"
adb uninstall com.ritikji.clientEcommerce  # only if an older build with another key is installed
adb install app/build/outputs/apk/release/app-release.apk
```

Android phones don’t appear in Finder

– that is normal on macOS (Android uses MTP). adb works anyway. To copy files use the free app **OpenMTP**, or just send the APK over WhatsApp/Drive and open it on the phone.

8 Create a Google Play developer account

1. Go to **play.google.com/console** and sign in with the Google account that will own the app.
2. Choose **Personal** account (for an individual) or **Organisation** (needs a D-U-N-S number).
3. Pay the one-time **US \$25** fee.
4. Complete **identity verification** (government ID) and verify your contact email and phone. This can take a few days.
5. Complete **Android developer verification** if Play Console asks for it (left menu).



Play Console home of a new personal account - ready to "Create app".

Personal accounts created after Nov 2023

must run a **closed test with at least 12 testers for 14 days** before they can apply for Production access (section 14).

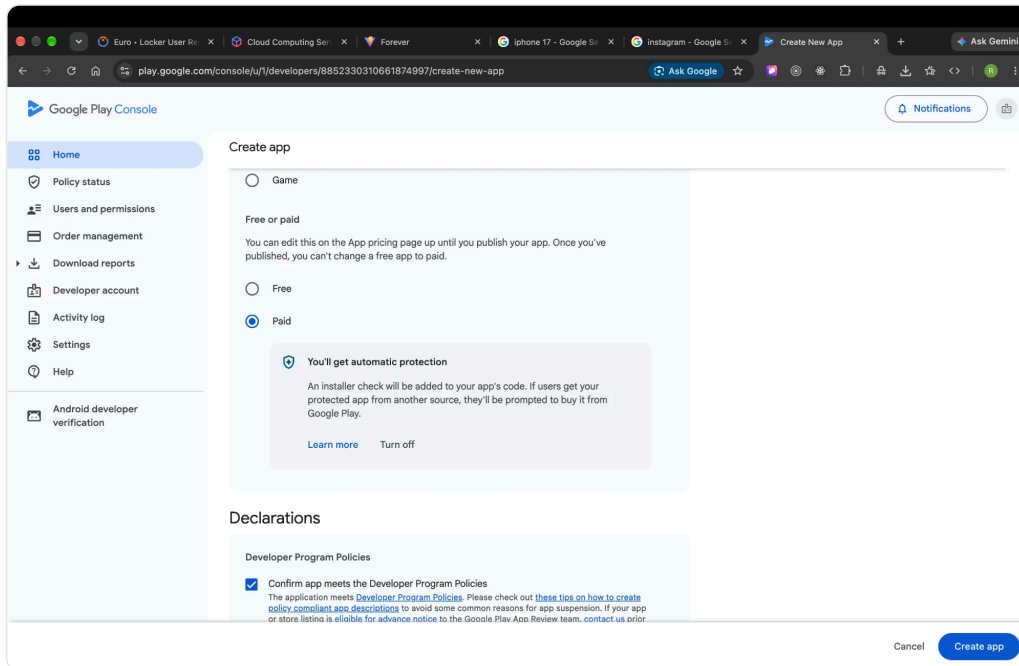
9 Create the app in Play Console

Home → **Create app**, then fill in:

Field	Value for Shopvra
App name	Shopvra (max 30 characters, can be changed later)
Package name	<code>com.ritikji.clientEcommerce</code> → click Check availability . Must match the AAB exactly. Permanent .
Default language	English (United States) - or English (India)
App or game	App
Free or paid	Free - users pay for products at checkout, not for the app
Declarations	Tick Developer Program Policies + US export laws

The screenshot shows the Google Play Console interface for creating a new app. The left sidebar contains navigation links: Home, Policy status, Users and permissions, Order management, Download reports, Developer account, Activity log, Settings, Help, and Android developer verification. The main content area is titled 'Create app' and 'App details'. It includes input fields for 'App name' (Shopvra) and 'Package name' (com.ritikji.clientEcommerce). Below the package name, a green checkmark and the text 'Package name available' are displayed. The 'Default language' is set to 'English (United States) - en-US'. Under 'App or game', the 'App' radio button is selected. At the bottom right, there are 'Cancel' and 'Create app' buttons.

Package name matches the app and shows “Package name available”.



Careful here: this screenshot shows **Paid** selected – for Shopvra choose **Free**.

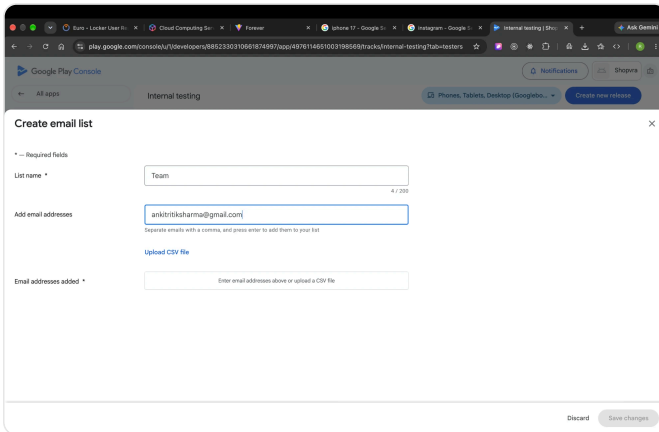
Selected “Paid” by mistake?

Before the first public release you can switch: **Monetize with Play** → **Products** → **App pricing** → **Change to free**. Once published as free, an app can never become paid.

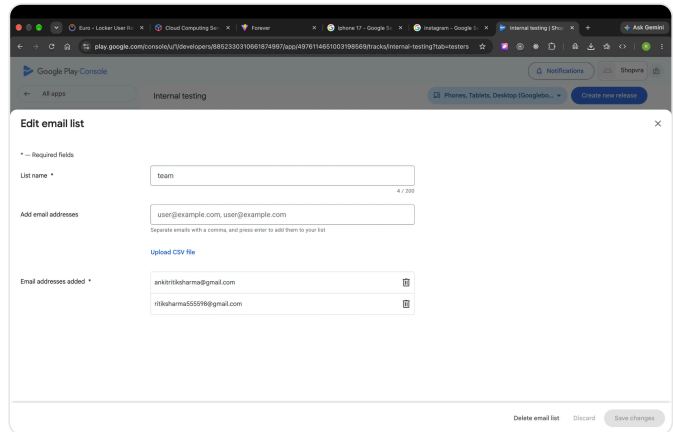
10 Add internal testers

Test and release → Testing → Internal testing → Testers tab → Create email list.

1. List name: e.g. **team**.
2. Type an email and **press Enter** - it must move into “Email addresses added”. Until then **Save changes** stays grey.
3. Click **Save changes**, then on the Testers tab make sure the list is **ticked** and click **Save**.



Email typed but not added yet → Save is disabled.



After pressing Enter the emails are listed → Save works.

Use the Play Store account of the phone

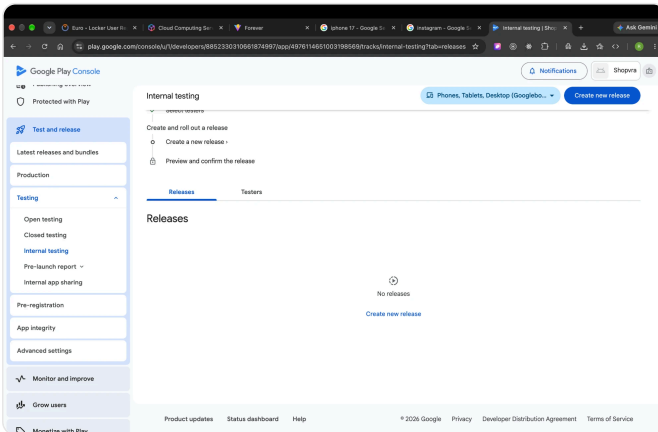
The tester email must be the Google account signed in to the Play Store on the test phone - otherwise the opt-in link says the app is not available. Internal testing allows up to 100 testers.

11 Upload the .aab and publish to internal testing

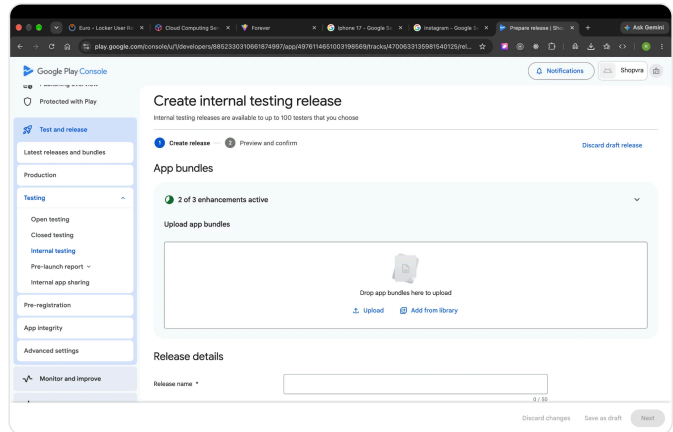
1. **Internal testing** → **Releases** → **Create new release**.
2. If asked about **Play App Signing**, keep **Use Google-generated key** (recommended).
3. **Upload** → choose `app-release.aab`.
4. Release name: **1 (1.0.0)** is suggested automatically. Release notes, e.g.:

```
<en-US>
First release of Shopvra: browse products, wishlist, cart, checkout (cash or card) and order tracking.
</en-US>
```

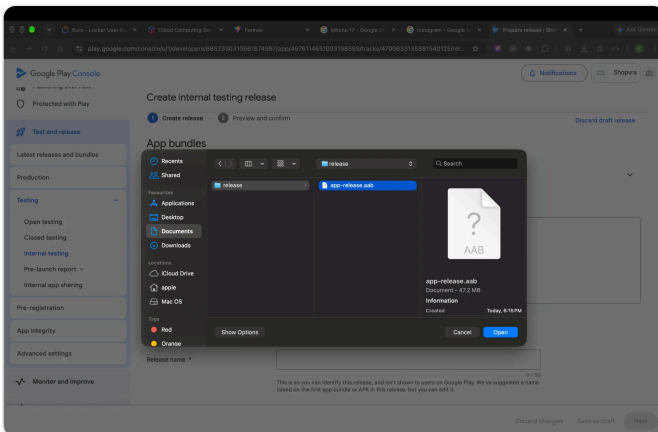
5. **Next** → review → **Save and publish**.



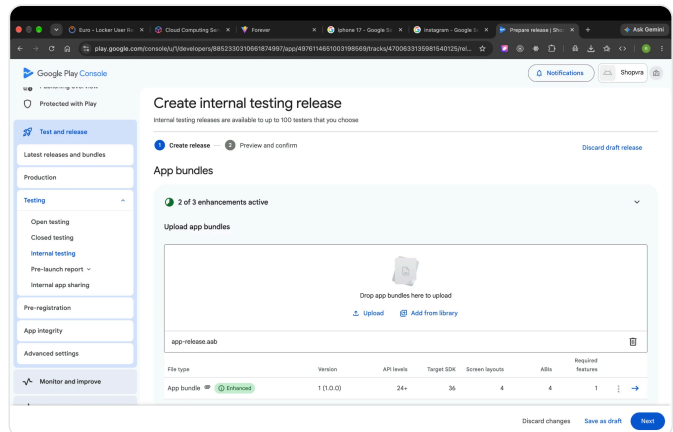
Internal testing - no releases yet.



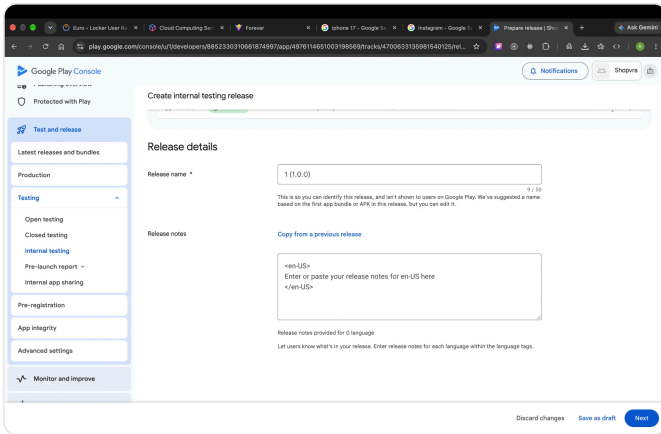
Create release - upload area.



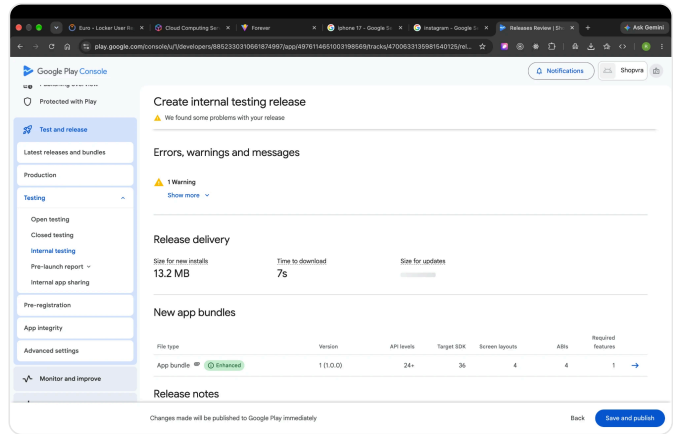
Pick `app-release.aab` (47.2 MB).



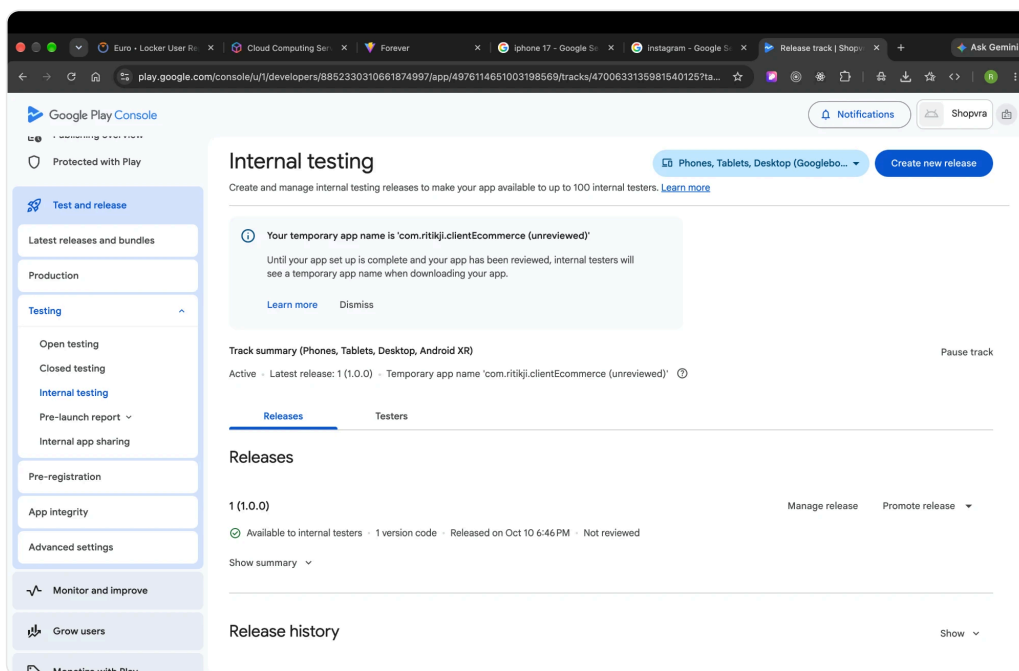
Bundle accepted: version 1 (1.0.0), API 24+, target SDK 36.



Release name and release notes.



Review: 13.2 MB download size, 1 warning (usually “no deobfuscation file” – harmless).



Published: “Available to internal testers”. The temporary name “com.ritikji.clientEcommerce (unreviewed)” switches to **Shopvra** after the store listing is reviewed.

12 Install from Google Play as a tester

1. **Internal testing** → **Testers** tab → **How testers join your test** → **Copy link**.
2. Open the link on the phone (Chrome, same Google account as the tester list) → **Accept invite**.
3. Tap **Download it on Google Play** → **Install**.

If the Play install fails

Uninstall the copy installed by USB first (it may be signed with a different key). A new release can take from a few minutes up to a couple of hours to reach testers.

13 Go-live roadmap after internal testing

Internal testing proves the build installs and runs. To publish Shopvra for everyone, finish these stages in order:

#	Stage	Where	Time
A	Deploy backend + website changes	AWS Console (Lambda, S3, CloudFront)	15 min
B	Upload release 1.0.1 to internal testing	Play Console → Internal testing	10 min
C	Complete App content declarations	Play Console → Policy → App content	30-45 min
D	Complete the store listing	Play Console → Grow users → Store presence	30 min
E	Closed testing with ≥ 12 testers for 14 days	Play Console → Closed testing	14+ days
F	Apply for production access	Play Console → Dashboard	~7 days review
G	Production release (staged rollout)	Play Console → Production	1-7 days review
H	After launch: monitor, reply, update	Play Console → Monitor and improve	ongoing

What changed in 1.0.1

Required by Google Play for apps with sign-up: **Profile** → **Delete account** (password-confirmed), a **Privacy Policy** link in the app, the web pages `shopvra.space/privacy` and `shopvra.space/delete-account`, and the backend endpoint `DELETE /api/auth/me`.

Stage A - Deploy the backend (AWS Lambda)

```
cd backend/expressJs/expressJs_ecommerce_backend
npm run build:lambda          # creates lambda.zip (~800 KB)
```

1. AWS Console (the account that hosts Shopvra) → **Lambda** → your Shopvra API function.
2. **Code** → **Upload from** → **.zip file** → choose `lambda.zip` → **Save**. Handler stays `index.handler`, runtime Node.js 22.x.
3. Test: `curl -X DELETE https://api.shopvra.space/api/auth/me` must answer `{"success":false,"message":"Not authorized, no token"}` (not 404).

Stage A - Deploy the website (S3 + CloudFront)

```
cd frontend/webapp_ecommerce_app
npm run build          # creates dist/
```

1. AWS Console → **S3** → the `shopvra.space` bucket → delete the old `assets/` folder → **Upload** everything inside `dist/` (not the folder itself).
2. **CloudFront** → the shopvra.space distribution → **Invalidations** → **Create** → path `/*`.
3. Check `https://shopvra.space/privacy` and `https://shopvra.space/delete-account` open (also in a private window).

Stage B - Upload 1.0.1

1. Version in `android/app/build.gradle` is already `versionCode 2`, `versionName "1.0.1"`.
2. Build: `cd frontend/mobile_ecommerce_app/cli/android && ./gradlew assembleRelease bundleRelease`
3. Play Console → **Internal testing** → **Create new release** → upload the new `app-release.aab` → notes “Account deletion and privacy policy link.” → **Save and publish**.
4. Testers get the update through the Play Store automatically.

14 App content declarations (exact answers)

Play Console → **Policy and programs** → **App content** (also linked from Dashboard → “Set up your app”).

Declaration	Answer for Shopvra
Privacy policy	<code>https://shopvra.space/privacy</code>
App access	All or some functionality is restricted → Add instructions: name “Reviewer account”, username + password of a dedicated demo user (sign up once in the app, e.g. <code>reviewer.shopvra@gmail.com</code>), note: “Sign in from Profile → Login. Browse, add to cart and checkout with Cash on Delivery.” Never give the admin account.
Ads	No , my app does not contain ads
Content rating	Start questionnaire → email → category All other app types → answer No to violence, sexuality, language, controlled substances, gambling, user interaction/sharing, location sharing. Answer Yes only to purchasing physical goods if asked. Expected rating: Everyone / 3+.
Target audience	18 and over only. “Could the app unintentionally appeal to children?” → No .
News app	No
Advertising ID	No - the app does not use the advertising ID (no <code>AD_ID</code> permission)
Government app	No
Financial features	My app doesn't provide any financial features (it sells physical goods; Stripe processes cards)
Health	My app does not have any health features
Data safety	See the table on the next page

Permissions are already clean

The 1.0.1 APK only requests `INTERNET` (+ biometric used by secure storage). No location, camera, contacts or media permissions - so no extra permission declarations are needed.

Data safety form

Question	Answer
Does your app collect or share any of the required user data types?	Yes
Is all of the user data collected by your app encrypted in transit?	Yes (HTTPS only)
Which account creation methods does your app support?	Username and password (email + password)
Delete account URL	<code>https://shopvra.space/delete-account</code>
Do you provide a way for users to request that some or all of their data is deleted without deleting the account?	No (they can remove addresses, cart and wishlist items themselves)

Data type	Collected	Shared	Purpose · required?
Personal info → Name	Yes	No	App functionality, Account management · Required
Personal info → Email address	Yes	No	App functionality, Account management · Required
Personal info → Address	Yes	No	App functionality (delivery) · Required to order
Financial info → Purchase history	Yes	No	App functionality · Required
App activity → Other actions (cart, wishlist)	Yes	No	App functionality · Optional
Everything else (location, contacts, photos, device IDs, payment card numbers...)	No	No	Card details are entered on Stripe's own checkout page, not collected by the app

“Shared” means given to another company for its own use. Service providers that process data for you (AWS, MongoDB Atlas, Stripe, Cloudinary) do not count as sharing.

15 Store listing (ready-to-paste text)

Grow users → Store presence → Main store listing. App category: **Shopping**. Tags: Shopping, Fashion.

Field	Text
App name	Shopvra
Short description (max 80)	Shop men, women & kids fashion, shoes and bags with fast, easy delivery.
Full description (max 4000)	Shopvra is your everyday fashion store - the latest clothing for men, women and kids, plus shoes and bags, in one simple app. Why Shopvra • Fresh collections and best sellers, updated regularly • Browse by category, search and filter by price • Save favourites to your wishlist • Pick your size and add to cart in one tap • Pay securely by card (Stripe) or choose Cash on Delivery • Save your delivery address for faster checkout • Track every order from placed to delivered • Your cart and wishlist sync between the app and shopvra.space Privacy first No ads and no tracking. Delete your account any time from your profile. Questions? Write to support@shopvra.space - we're happy to help.
Contact email	support@shopvra.space (must be a working inbox)
Website	https://shopvra.space
Graphic	Spec
App icon	512 × 512 PNG, 32-bit, ≤ 1 MB - export from <code>cli/src/assets/images/brand/shopvra-icon-square.svg</code>
Feature graphic	1024 × 500 JPG/PNG, no transparency - logo + tagline on the brand black/coral
Phone screenshots	2-8 images, portrait 9:16 (e.g. 1080 × 1920): Home, Shop, Product, Cart, Checkout, Orders

16 Closed testing (≥ 12 testers $\times$ 14 days)

1. **Test and release** → **Testing** → **Closed testing** → **Manage track** (the default track is usually named “Closed testing - Alpha”).
2. **Countries / regions** tab → add India (and any other countries).
3. **Testers** tab → choose **Email lists** (create a list with at least **12 Gmail addresses**) or a **Google Group** (easier: create *shopvra-testers@googlegroups.com* and let people join). Add a **feedback URL or email**. **Save**.
4. **Releases** → **Create new release** → **Add from library** → pick version **2 (1.0.1)** (no new upload needed) → **Next** → **Save and publish**. Or from Internal testing use **Promote release** → **Closed testing**.
5. The first closed release goes to **Google review** (hours to a few days). After approval, copy the **opt-in link** from the Testers tab and send it to testers.
6. Each tester: open the link → **Become a tester** → install from Play. Ask them to use the app for a few minutes on several days.
7. Keep **at least 12 opted-in testers continuously for 14 days**. The Dashboard shows a progress counter.

Common reasons the 14 days restart or don't count

Fewer than 12 testers opted in at some point, testers who joined but never installed, or the track being paused. Recruit 15-20 people to be safe.

17 Apply for production access

1. When the counter reaches 14 days: **Dashboard** → **Apply for production**.
2. Answer the form honestly: how you recruited testers, how engaged they were, what feedback you got and what you changed, who the app is for (customers of shopvra.space) and how you know it is ready (tested ordering, payments, account deletion...).
3. Google replies by email, usually within **7 days**. If rejected, keep testing and apply again.

18 Production release

1. Make sure the Dashboard shows every “Set up your app” task completed (green).
2. **Test and release** → **Production** → **Countries / regions** → add the countries where you deliver.
3. **Create new release** → **Add from library** (or promote from closed testing) → release notes for users, e.g.

```
<en-US>
Welcome to Shopvra! Shop the latest fashion, save favourites, pay by card or cash on delivery, and track your
orders.
</en-US>
```

4. **Staged rollout**: start at **20%** of users. Check **Android vitals** (crashes/ANRs) for 1-2 days, then **Manage release** → **Update rollout** to 50% → 100%. If something breaks, **Halt rollout**.
5. **Next** → **Save** → **Send for review** (Publishing overview → **Send changes for review**). First reviews take 1-7 days.
6. When approved, the app is live at `https://play.google.com/store/apps/details?id=com.ritikji.clientEcommerce`. Use **Publishing overview** → **Managed publishing** if you want to choose the exact launch moment.

19 After launch

- **Monitor and improve** → **Android vitals**: keep crash rate < 1.09% and ANR rate < 0.47% (Google’s bad-behaviour thresholds).
- **Ratings and reviews**: reply to reviews; fix what users report.
- **Policy status** and the Play Console inbox: answer any policy emails quickly (deadlines apply).
- Keep the backend (`api.shopvra.space`) and website up - the store listing links to them.
- Google requires apps to target a recent Android API level each year (currently 36) - update React Native / `targetSdkVersion` when asked.
- Ship updates with a higher `versionCode` every time (next section).

20 Releasing an update later

1. Make and test the code changes (`npm run validate`).
2. In `android/app/build.gradle` increase the version - **versionCode must always go up**:

```
versionCode 3          # was 2 - Play rejects an AAB with a code it has already seen
versionName "1.0.2"    # what users see
```

3. Build and verify (sections 5-6), using the **same upload key**.
4. Play Console → the track (internal / closed / production) → **Create new release** → upload → notes → publish.

Change type	Example versionName	versionCode
Bug fix	1.0.0 → 1.0.1	+1
New feature	1.0.1 → 1.1.0	+1
Big redesign	1.1.0 → 2.0.0	+1

21 Troubleshooting

Problem	Fix
<code>bundleRelease</code> needs your upload key...	The 4 <code>MYAPP_UPLOAD_*</code> lines are missing from <code>~/gradle/gradle.properties</code> (section 4).
Keystore was tampered with, or password was incorrect	Wrong <code>MYAPP_UPLOAD_STORE_PASSWORD</code> / <code>KEY_PASSWORD</code> .
Build fails: <i>react-native-worklets ... prefab ... not readable</i>	You ran <code>clean</code> together with the build - run <code>./gradlew clean</code> separately (section 5).
Play: "You uploaded an APK or Android App Bundle that was signed in debug mode"	The AAB was signed with the debug key - fix section 4 and rebuild.
Play: "Version code 1 has already been used"	Increase <code>versionCode</code> (section 15).
Play: "Upload certificate doesn't match"	The AAB was signed with a different keystore - always use <code>shopvra-upload.keystore</code> .
<code>adb devices</code> shows nothing	Enable USB debugging (+ Install via USB on Xiaomi), set USB mode to File transfer, accept the RSA prompt, try another data cable.
<code>adb install</code> fails / <code>INSTALL_FAILED_UPDATE_INCOMPATIBLE</code>	The installed app has another signature - <code>adb uninstall com.ritikji.clientEcommerce</code> first.
Email list "Save changes" is grey	Press Enter after typing each email so it is added to the list.
Tester link says app not available	Phone's Play Store account isn't on the tester list, the list isn't ticked, or the release is still propagating.
App created as Paid	Monetize with Play → Products → App pricing → Change to free (only before publishing).

22 Backup & security checklist

- ✓ Copy `~/keys/shopvra-upload.keystore` to **two** safe places (e.g. encrypted cloud drive + USB drive).
- ✓ Store the keystore password, alias (`upload`) and the SHA-256 above in a **password manager**.
- ✓ Never commit the keystore or `gradle.properties` with passwords to git; never paste the password in chats or screenshots.
- ✓ Keep Play App Signing enabled - if the upload key is lost or leaked, request an upload key reset in **Play Console → Test and release → App integrity → App signing**.
- ✓ Use a separate demo account for Google reviewers; never share the admin login.
- ✓ Turn on 2-step verification for the Google account that owns the Play Console.

